

Economic and Social Modulations of Innate Decision-Making in Mice Exposed to Visual Threats

Reviewed Preprint

v1 • July 28, 2025

Not revised

Zhe Li, Jiahui Wang, Yidan Sun, Jialin Li, Ling-yun Li, Ya-tang Li 

Academy for Advanced Interdisciplinary Studies, Peking University, Beijing, China • Beijing Institute for Brain Research, Chinese Academy of Medical Sciences & Peking Union Medical College, Beijing, China • Chinese Institute for Brain Research, Beijing, China • Department of Neurobiology, School of Basic Medical Sciences, Capital Medical University, Beijing, China

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

eLife Assessment

The authors show that innate defensive behavior in mice is shaped by threat intensity, reward value, and social hierarchy, highlighting how value and social context influence instinctive decisions. The authors provide **useful** behavioural findings supported by strong data, yet the evidence is **incomplete** due to ambiguities about methodology and the computational model that remains largely descriptive.

<https://doi.org/10.7554/eLife.107306.1.sa3>

Abstract

When confronted by a predator, most animals make innate decisions with rapid reaction times – a trait shaped by natural selection to maximize survival. However, in complex and dynamic environments, fast reactions are meaningful only when grounded in accurate judgments and correct choices, which often require cognitive control. Here, we investigate how threat intensity, reward value, and social hierarchy influence behavioral decisions in foraging mice exposed to overhead visual threats. We find that threat intensity plays a dominant role in decision-making: elevated threat levels trigger robust defensive responses, including shorter latencies to flee, increased fleeing speeds, and longer escape distances. The influence of reward is context-dependent: at low threat levels, higher reward values suppress defensive responses, leading to prolonged time in the reward zone, slower speeds, and shorter escape distances. In contrast, under high-threat conditions, increasing reward value enhances sensitivity to threats, as evidenced by shorter latencies to flee and heightened vigilance. Social hierarchy further shapes decision-making, with dominant mice exhibiting greater vigilance and a stronger preference for risk-averse behaviors compared to subordinates. To quantify the decision-making process, we developed a drift-diffusion leaky integrator model that successfully captures how mice integrate threat intensity, reward value, and vigilance into their behavioral decisions. Our findings reveal the economic and social modulation of survival decisions, offering insights into the computational mechanisms underlying the interplay between instinctive reaction and cognitive control in innate decision-making.

1 Introduction

How animals make decisions among alternative actions is a key question in neuroscience. In natural environments, decisions emerge from the integration of sensory inputs, past experience, and internal states. Such innate decision-making requires no training, and animals learn quickly from experience (Meister, 2022 [↗](#)). However, in the laboratory, most decision-making researchers rely on trained behaviors such as two-alternative forced choice (2AFC) tasks, where animals undergo weeks of training involving thousands of trials (Burgess et al., 2017 [↗](#)). This overtraining can induce brain-wide synchronized neural activity (Steinmetz et al., 2019 [↗](#)), potentially engaging neural circuits distinct from those evolved for innate decision-making.

Innate decision-making plays an important role in animal survival and reproductive success. For example, a prey animal that decides not to escape a lethal predator dies. Evolutionary pressure has shaped these responses to occur within extremely short time windows to enhance the likelihood of survival. However, speed alone is not enough – these decisions must also be correct. In complex and ever-changing environments, correct decisions often require cognitive regulation to assess risk, evaluate alternative defensive strategies, and adjust responses accordingly (Evans et al., 2019 [↗](#); Ydenberg and Dill, 1986 [↗](#)).

Supporting this notion, studies across species have revealed how decisions to threats are influenced by various factors. Crayfish modify their reactions based on food availability (Liden et al., 2010 [↗](#)), satiation levels (Schadegg and Herberholz, 2017 [↗](#)), and social status (Krasne et al., 1997 [↗](#)). In mice, decisions to overhead visual threats depend on the physical properties of visual stimuli (De Franceschi et al., 2016 [↗](#); Liden and Herberholz, 2008 [↗](#); Tammero and Dickinson, 2002 [↗](#); Yang et al., 2020 [↗](#); Yilmaz and Meister, 2013 [↗](#)), learned knowledge from experiences (Vale et al., 2017 [↗](#)), and housing conditions (Lenzi et al., 2022 [↗](#)). However, despite mice being social animals and most predator encounters occurring during foraging, little is known about how they weigh perceived risks and rewards when making defensive decisions, or how these decisions are influenced by social hierarchy.

To address these questions, we developed an ecologically relevant behavioral paradigm examining decision-making in foraging mice exposed to overhead visual threats. Our findings demonstrate that defensive choices are collectively influenced by threat intensity, reward value, and social hierarchy. Threat intensity plays a dominant role in driving decisions, while the influence of reward depends on threat levels. Under low-threat conditions, elevated reward suppresses defensive responses, consistent with a value-based economic strategy. Yet the same reward increase enhances defensive responses during high-threat scenarios, indicating a shift to a survival-priority strategy. Furthermore, social hierarchy plays a significant role, with dominant mice exhibiting stronger risk-aversion behaviors and heightened vigilance. To quantify the contributions of various factors to decision-making, we proposed a drift-diffusion leaky integrator model that successfully captures key characteristics of the experimental results. Our work provides a valuable behavioral framework for future investigations into the neural mechanisms underlying cognitive control of innate decision-making.

2 Results

2.1 A behavioral paradigm for studying instinctive decision-making in mice

To investigate how animals make innate decisions in natural environments, we designed a behavioral paradigm to simulate the escape behavior of foraging mice in the wild. In this paradigm, a group of 2-5 co-housed mice were placed in a nest, with each individual identified using a radio frequency identification (RFID) tag. Only one mouse was allowed to enter a linear arena to get water delivered at the end. As the mouse approached the reward, an overhead expanding dark disc that mimics the approach of an aerial predator was triggered (**Figure 1A**), forcing the animal to decide whether to risk getting the reward or defend for safety. Behavioral data were recorded using a ground-mounted camera, and DeepLabCut was used to track the movements of the mouse's nose and tail base (Mathis et al., 2018) (**Figure S1A**). Compared to the exploratory phase, the presence of looming stimuli significantly increased the frequency of arena entries and decreased the duration of each visit (**Figures 1B** and **1C**).

To identify distinct behavioral patterns in response to the looming stimuli, we defined 19 behavioral features from key body points of the mouse and fed them into a random forest classifier to predict its decisions (**Figures 1D** and **S1B**, see Methods). Animal decisions across 3861 trials were categorized into four types: direct flight (11.8%), flight after assessment (43.2%), freezing (5.8%), and no reaction (39.2%, **Figures 1E** and **1F**). The classification model achieved an accuracy of 0.95 (**Figure S1C**). A key distinction between “assessment+flight” and “freezing” decisions was the duration of stationary behavior: mice that decided to freeze remained stationary significantly longer than those in the “assessment+flight” group (**Figure 1G**). Furthermore, the latency to flee differed significantly between “direct flight” and “assessment+flight” decisions (**Figure 1H**).

We hypothesized that fear level increases progressively across the decision types: “freezing”, “flight after assessment”, and “direct flight”. This is supported by the observations that mice in the “direct flight” group exhibited higher flee speeds (**Figure 1I**) and shorter hiding latency (**Figure 1J**). Additionally, the proportion of mice that recovered within 20 seconds of the stimulus was about 20%, 40%, and 80% for “direct flight”, “assessment+flight”, and “freezing”, respectively (**Figure 1K**). Consistently, recovery time decreased progressively across these decision types (**Figure 1L**). Our findings align with the theory of predatory imminence continuum (Fanselow and Lester, 1988), which proposes that defensive responses become more intense as the perceived threat level increases.

Although the looming stimulus triggered robust defensive behavior, animals quickly learned to recognize it as non-threatening (Lenzi et al., 2022; Vale et al., 2017). Our behavioral data showed a gradual habituation to the looming stimulus, beginning immediately after the first trial (**Figure 2A**). Specifically, the defensive probability decreased substantially within the first few trials (**Figures 2B** and **2C**), indicating rapid adaptation to the stimulus. Hiding latency increased progressively across trials (**Figure 2D**), stabilizing after the 12th trial. Intriguingly, peak fleeing speed increased over the first five trials before declining (**Figure 2F**), suggesting a rapid modulation of vigilance after just a few exposures to the threat.

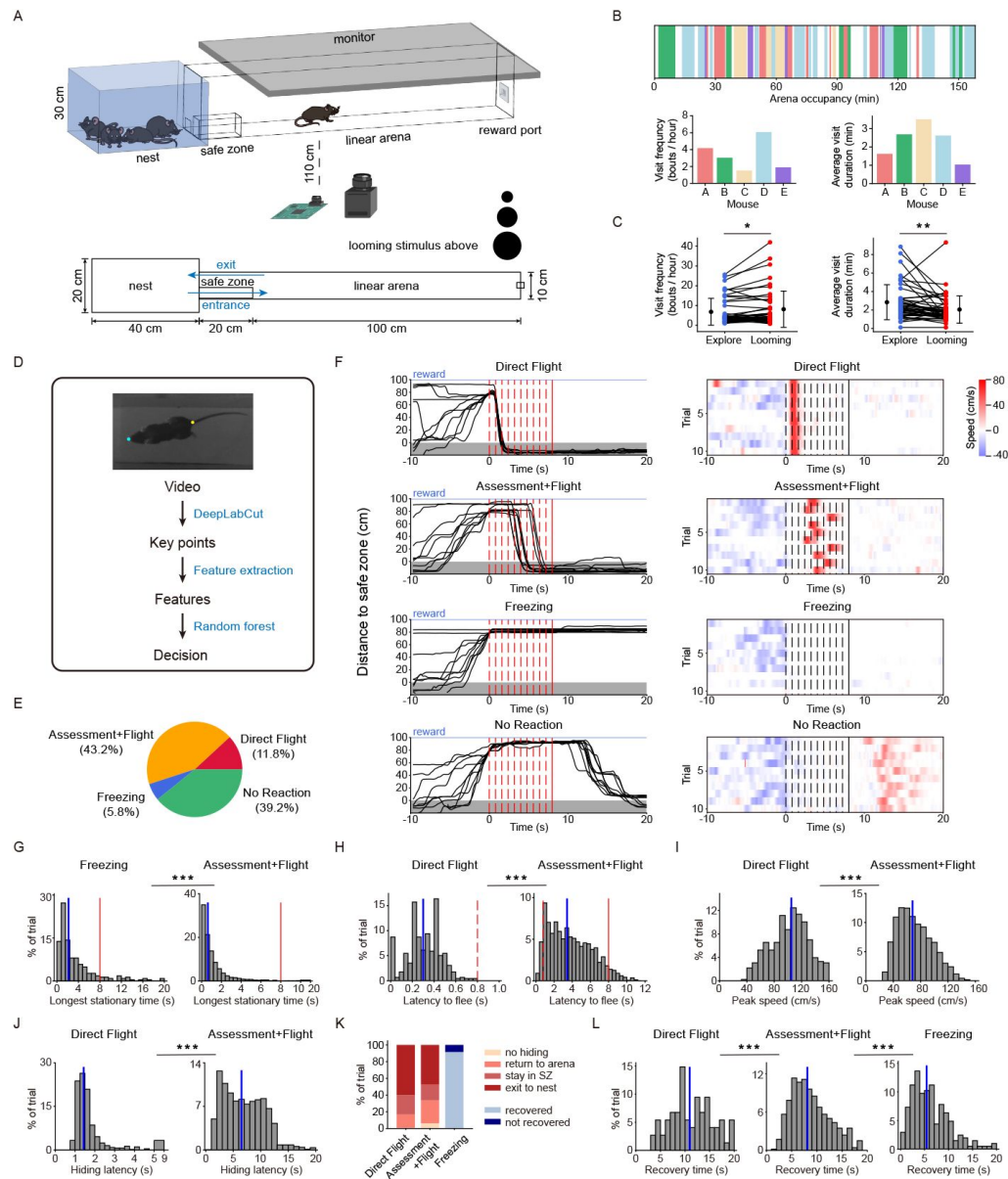


Figure 1

A behavioral paradigm for investigating innate decision-making in mice.

(A) Schematic of the behavioral assay (3D and top-down views). (B) Top: arena occupancy patterns for five example mice in one session. Bottom: visit frequency and average duration for each visit. Colors mark the mouse's identity. (C) Visit frequency and duration under exploration and looming conditions. Error bars represent the standard deviation. $N = 46$ mice, paired t -test. (D) The pipeline for behavioral classification. (E) Distribution of behavioral decisions across 3862 trials from 140 mice. (F) Left: distance to the safe zone for four decision types (10 example trials each). Red dashed lines mark the onset of each stimulus repetition; solid lines mark the end of the last repetition. Grey shade indicates the safe zone. Right: locomotion speed towards the safe zone for the same trials. (G) Distribution of the longest stationary time for "freezing" and "assessment+flight". $N = 224$ and 1667 , $p < 0.001$, Mann-Whitney-Wilcoxon test. (H) Distribution of latency to flee for "direct flight" and "assessment+flight". $N = 458$ and 1667 , $p < 0.001$, Mann-Whitney-Wilcoxon test. (I) Distribution of peak speed for "direct flight" and "assessment+flight". $N = 458$ and 1667 , $p < 0.001$, Mann-Whitney-Wilcoxon test. (J) Distribution of latency to hide in the safe zone for "direct flight" and "assessment+flight". $N = 455$ and 1563 , $p < 0.001$, Mann-Whitney-Wilcoxon test. (K) Distribution of behavioral states at the end of each trial for "direct flight", "assessment+flight", and "freezing". (L) Distribution of fear recovery time for "direct flight", "assessment+flight", and "freezing". $N = 74$, 458 , and 205 , $p < 0.001$, Kruskal-Wallis test and Dunn's post-hoc test.

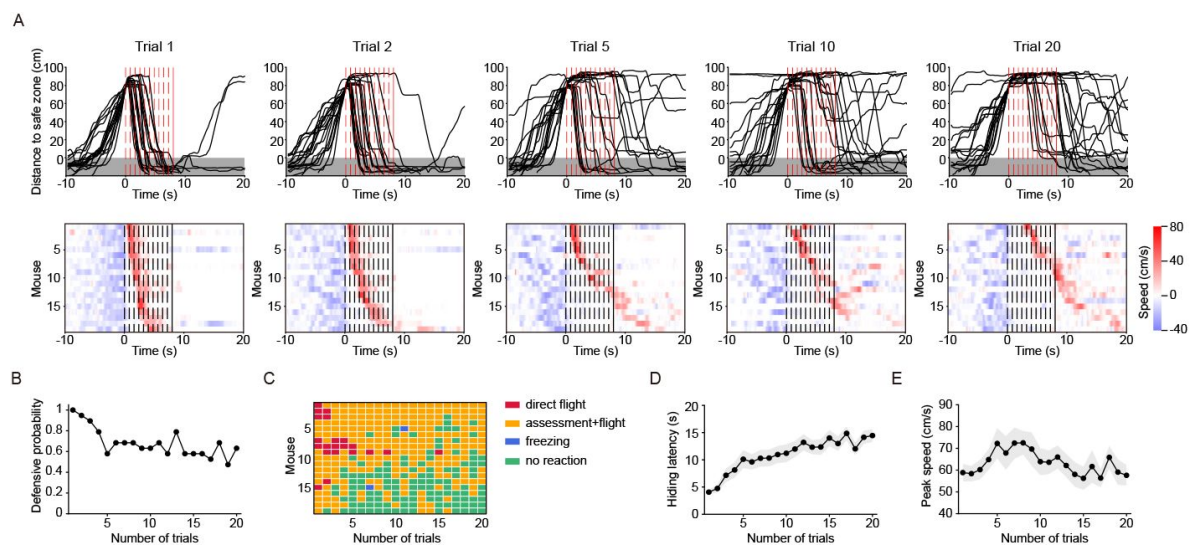


Figure 2

Mice learn quickly from experience.

(A) Distance to the safe zone (top) and locomotion speed (bottom) for the 1st, 2nd, 5th, 10th, and 20th trials of 19 mice. Dash lines mark the start of each stimulus, and solid lines mark the end of the stimulus. Positive speed is towards the safe zone. (B) Defensive probability decreases with the number of trials. (C) Summary of the decisions made by 19 mice for the first 20 trials. (D) Latency to flee increases with the number of trials. Gray shade indicates the standard error of the mean. (E) Hiding latency increases with the number of trials. N=19. (F) Peak fleeing speed varies with the number of trials. N=19.

2.2 Mice make economic decisions modulated by vigilance

In the wild, most prey-predator encounters occur while prey animals are foraging. It remains unclear how prey balance the value of food against the risk of predation in their decision-making process. Recent work has shown that rodents make economically driven decisions in learned tasks that require intensive training (Constantinople et al., 2019 [DOI](#)). However, it is unknown whether these findings extend to natural behaviors. Furthermore, prey animals have evolved to maintain high levels of vigilance during foraging to increase their chances of survival. Yet, the interplay among food value, threat intensity, and vigilance in shaping defensive decisions remains elusive.

Here, we demonstrate that mice exhibit distinct behavioral patterns in response to varying threat intensities and reward values (**Figure 3A** [DOI](#)). As threat intensity increased, animals were more likely to prioritize threat avoidance and engage in direct flight behaviors (**Figure 3B** [DOI](#)), indicating that perceived threat plays a dominant role in their decision-making processes. This was further evidenced by longer flight distances (**Figure 3C** [DOI](#)) and reduced time spent in the reward zone (**Figure 3D** [DOI](#)), suggesting a trade-off between threat avoidance and reward pursuit. Mice also fled more quickly under higher threat levels (**Figure 3E** [DOI](#)). This reduction in reaction time, a measure of vigilance (Buck, 1966 [DOI](#)), suggests heightened vigilance in response to increased threat. This is consistent with findings in birds, where elevated predation risk increases vigilance and reduces feeding time (Caraco et al., 1980 [DOI](#)). Furthermore, fleeing speeds increased significantly with threat level (**Figure 3F** [DOI](#)), reflecting the combined effects of perceived higher risk and heightened vigilance.

While threat intensity plays a major role in decision-making, the influence of reward is more nuanced and depends on the threat context. Under low-threat conditions, decisions were primarily driven by perceived reward value. We first confirmed that sucrose was more valuable than water by measuring consumption during exploration, which remained unaffected by low-level threats (**Figure 3G** [DOI](#)). As reward value increased, mice exhibited fewer defensive responses (**Figure 3B** [DOI](#)), indicating that they weighed risk and reward to make economic decisions. Higher rewards also led to shorter flight distance (**Figure 3C** [DOI](#)) and longer stays in the reward zone (**Figure 3D** [DOI](#)), supporting the value-based decision-making strategy. Interestingly, reward value had little effect on latency to flee (**Figure 3E** [DOI](#)), suggesting that vigilance remains relatively stable across reward conditions. Consistently, fleeing speed decreased as reward value increased (**Figure 3F** [DOI](#)).

Conversely, under high-threat conditions, increasing reward value led to more direct flight behaviors (**Figure 3B** [DOI](#)). This counterintuitive observation can be explained by heightened vigilance induced by the increased reward value. The positive correlation between reward value and vigilance likely reflects evolutionary pressures, where high risk is often associated with high reward. In these scenarios, escaping the predator becomes the top priority, even in the presence of high rewards. Evidence for increased vigilance includes shorter latencies to flee (**Figure 3E** [DOI](#)) and higher escape speeds (**Figure 3F** [DOI](#)) when a reward was present. Further supporting this, mice spent more time monitoring the upper visual field during the second trial while approaching sucrose (**Figure 3H** [DOI](#)). These results suggest that, under high-threat conditions, reward value indirectly shapes decision-making through its impact on vigilance.

To confirm value-based decision-making under low-threat conditions while controlling for individual variation, we performed behavioral experiments using the same mice (**Figures 3I** [DOI](#) and **3J** [DOI](#), see Methods). When a reward was introduced, mice showed reduced defensive probabilities (**Figures 3K** [DOI](#) and **3L** [DOI](#)), increased duration in the reward zone, and slower escape speeds (**Figure 3M** [DOI](#)). Latency to flee remained unaffected (**Figure 3M** [DOI](#)), reinforcing the idea

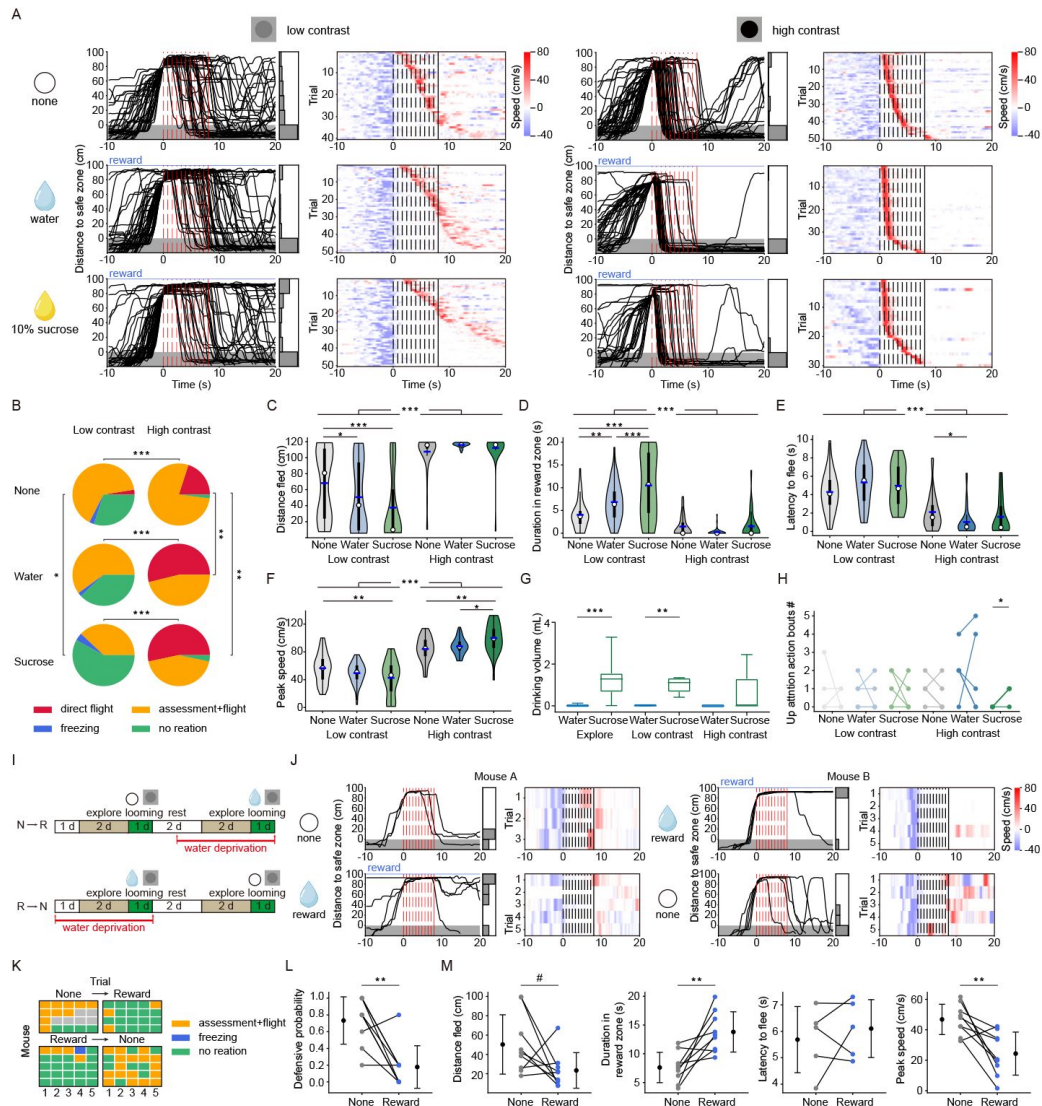


Figure 3

Mice make economic decisions modulated by vigilance.

(A) Distance to the safe zone and locomotion speed towards the safe zone across trials in response to low- and high-contrast looming stimuli with different reward values. Dash lines indicate stimulus onset; solid lines mark stimulus offset. (B) Distribution of decision types for six experimental conditions. $N = 40$ trials from 4 mice (none, low), 50 trials from 5 mice (water, low), 50 trials from 5 mice (sucrose, low), 50 trials from 5 mice (none, high), 39 trials from 5 mice (water, high) and 30 trials from 5 mice (sucrose, high), Chi-squared test. (C-F) Distance fled, duration in the reward zone, latency to flee, and peak fleeing speed across all conditions. Two-way ANOVA with post hoc Tukey's test. $N = 40, 50, 50, 50, 39, 30$ trials for C, D, F, and $N = 27, 30, 19, 49, 39, 29$ trials for E. (G) Water or sucrose consumption during exploration and looming experiments at low and high contrasts. $N = 10, 10, 5, 5, 5, 5$ trials. Mann-Whitney-Wilcoxon test. Whiskers indicate the minimum and maximum. Boxes indicate the first and third quartiles. (H) Number of up-attention action bouts before exposed to looming at the first and second trials. $N = 4, 5, 5, 5, 5, 5$ mice. Wilcoxon signed-rank test. (I) Experimental timeline investigating the influence of water reward on innate decision-making in the same mouse. (J) Distance to the safe zone and locomotion speed towards the safe zone across trials in response to low-contrast looming stimuli with and without water reward in an example mouse. Left, non-reward condition followed by reward condition; Right, reward condition followed by non-reward condition. (K) Decision types of nine mice across five trials. Gray squares indicate trials where the mouse did not enter the arena within 30 min. (L) Defensive probability in non-reward and reward conditions. $N = 9$ mice, paired t -test. (M) Distance fled, duration in reward zone, latency to flee, and peak speed in non-reward and reward conditions. $N = 5$ mice for latency to flee and 9 mice for other features, paired t -test, two-sided. For all panels: $\#p < 0.1$, $**p < 0.05$, $***p < 0.01$, $****p < 0.001$.

that vigilance is unaffected by reward under low-threat conditions. Collectively, these findings reveal how innate decision-making in response to looming stimuli is shaped by the dynamic interplay between perceived threat intensity, reward value, and vigilance.

2.3 Influence of social hierarchy structure on decision-making

Mice are social animals that live in groups, where social hierarchy often plays a critical role in shaping behavior. To investigate how social rank influences decision-making in response to looming threats, we compared the behaviors of dominant and subordinate mice. Rank order within each mouse pair was determined using the tube test before and after behavioral experiments (**Figure 4A** [↗](#), see Methods), and only pairs with stable rankings were included in the analysis.

We first examined how arena occupation was influenced by social rank before and during looming stimulation. Subordinate mice spent more time exploring the arena than dominant mice during the pre-looming exploration (**Figures 4B** [↗](#) and **4C** [↗](#)). However, this difference disappeared during and after exposure to the looming stimulus. No differences were observed in the number of arena entries (**Figure 4D** [↗](#)). These findings suggest that looming has a long-lasting impact on animal behavior.

Interestingly, when the threat co-localized with the reward, only dominant mice reduced their time in the reward zone (**Figures 4E** [↗](#) and **4F** [↗](#)), indicating greater vigilance compared to subordinates. This heightened vigilance in dominant mice was further supported by reduced habituation to the looming threat (**Figures 4G** [↗](#) and **4H** [↗](#)) and a higher proportion of defensive decisions (**Figure 4I** [↗](#)). Consistent with their heightened vigilance, dominant mice fled with shorter latencies and higher speeds (**Figure 4J** [↗](#)). Moreover, they prioritized threat avoidance over reward – fleeing longer distances and spending less time in the reward zone (**Figure 4J** [↗](#)).

These findings indicate that dominant mice prioritize threat avoidance over reward and are more sensitive to risk. In contrast, subordinates are more easily drawn to the allure of reward and display reduced vigilance in the presence of threats.

2.4 A model for innate decision-making

Our experimental results demonstrate that innate decision-making in response to visual threats is influenced by perceived threat intensity, reward value, and vigilance. Specifically, the distinct effects of reward under low- and high-threat conditions can be explained by taking the animal's vigilance level into account. To interpret these behavioral results quantitatively, we developed a model of innate decision-making, in which the evidence for escape was modeled as a drift-diffusion leaky integrator (see Methods). In this model, threat intensity and vigilance jointly impact the threat gain and charge the integrator, while a constant leak drives it back towards a stable state. Reward value modulates the drift rate, discharging the integrator and thereby reducing the likelihood of escape.

Model parameters were estimated in two steps. First, we fit the model to experimental data without considering reward and identified the optimal leakage rate, threat gain, diffusion rate, and escape threshold. This revealed distinct decision-making dynamics under low- and high-threat conditions. At low threat, the average evidence level remained below the escape threshold, and escape decisions emerged from stochastic fluctuations in individual trials (**Figure 5A** [↗](#)). At high threat, the average evidence level crossed the threshold, indicating that escape decisions were mainly driven by threat gain (**Figure 5B** [↗](#)). Notably, threat gain and diffusion rate had distinct effects on latency to flee: threat gain shifted the mean latency, while diffusion rate affected its variance. Thus, the model captured not only the observed decision patterns but also the distributions of latency to flee.

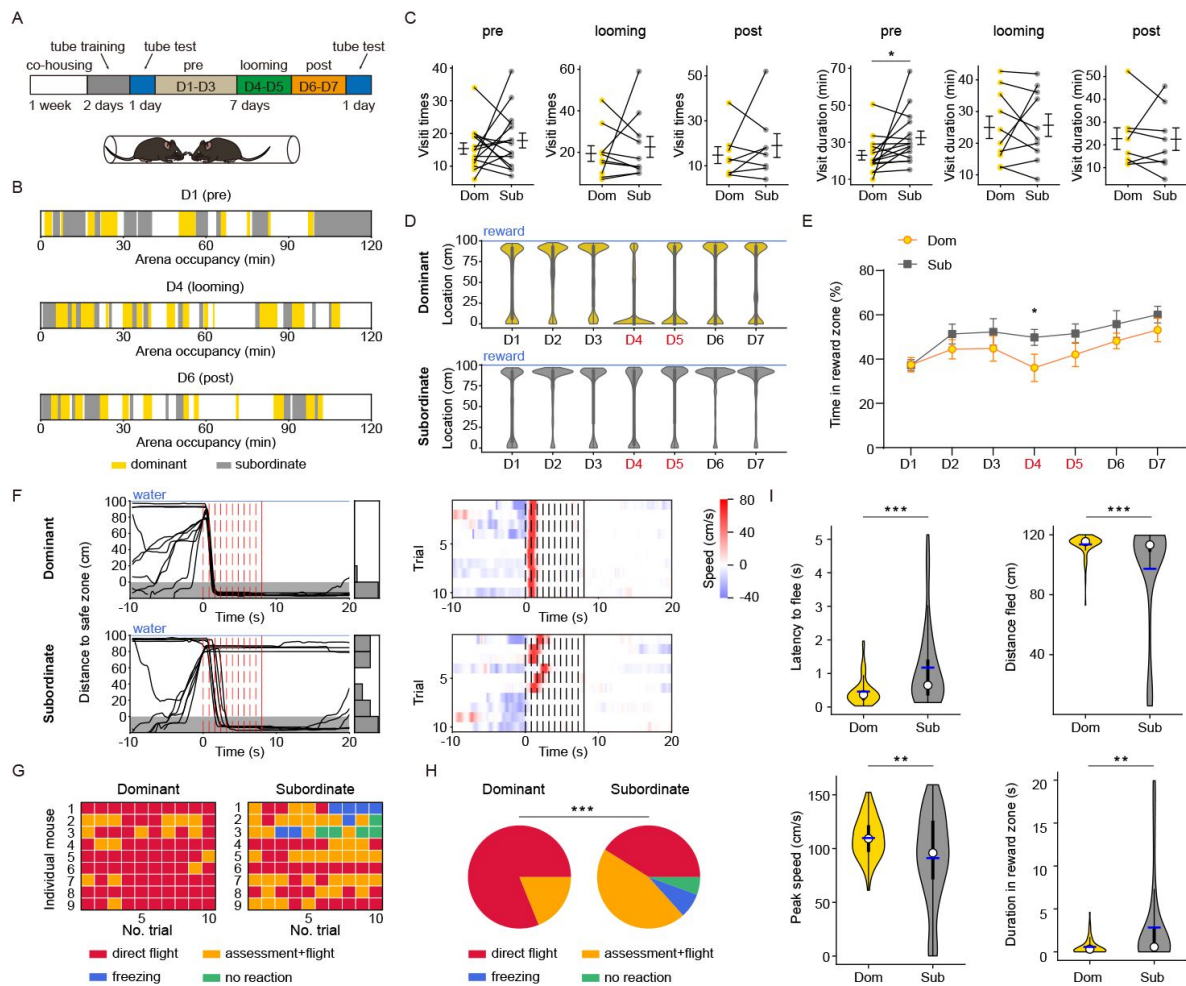


Figure 4

Influence of social hierarchy structure on innate decision-making.

(A) Schematic of the social hierarchy experiment. Top, experimental timeline. Each session lasted 2 hours per mouse pair during the pre-loom, looming, and post-loom phases. Bottom, schematic of the tube test. (B) Arena occupancy for an example pair of mice during the three sessions. (C) Visiting frequency and average duration per visit for dominant and subordinate mice during the pre-loom, looming, and post-loom sessions. N = 9 pairs, paired *t*-test. (D) Distance to the safe zone over seven days for an example pair. Looming stimuli were presented on days 4 and 5. (E) Percentage of time spent in the reward zone across days. Error bars represent SEM. N = 9 pairs, paired *t*-test. (F) Distance to the safe zone (left) and locomotion speed (right) for an example pair. (G) Behavioral decisions across the first 10 trials for 9 mouse pairs. (H) Pie charts showing decision distributions for dominant and subordinate mice. N = 90, 90 trials, Chi-squared test. (I) Violin plots comparing dominant and subordinate mice for: latency to flee (N = 90, 78 trials), distance fled, peak fleeing speed, and duration in the reward zone (N = 90, 90 trials) for dominant and subordinate mice. Mann-Whitney-Wilcoxon test. For all panels: ***p* < 0.05, ****p* < 0.001.

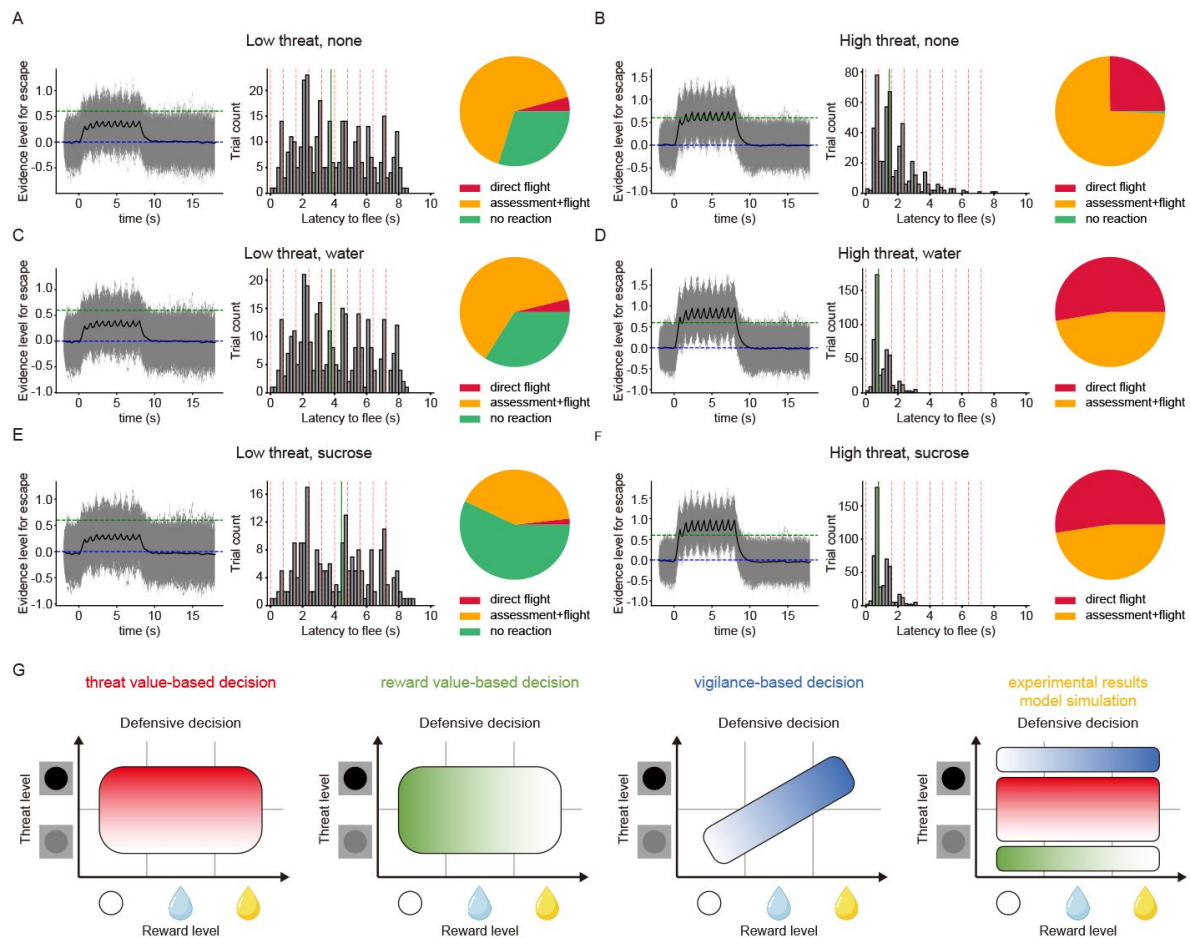


Figure 5

The drift-diffusion leaky integrator model.

(A-F) Simulated evidence level for escape and predicted latencies to flee and decision types across six conditions. (G) Compare decision models based on threat intensity, reward value, and vigilance to experimental results. Color saturation indicates the likelihood of defensive decisions.

In the second step, we incorporated the effect of reward value and identified the optimal drift rate and threat gain across different threat and reward conditions. Consistent with experimental results showing that decisions at low threat were primarily driven by reward value, drift rate varied across reward conditions (0, 0.01, and 0.05 for no reward, water, and sucrose, respectively), while threat gain remained relatively constant (1, 1, and 0.9 for no reward, water, and sucrose, respectively). Accordingly, the model reproduced the finding that increasing reward value reduced the likelihood of escape without affecting latency to flee (**Figures 5A, 5C**, and **5E**). The same drift rates were applied to model the decision-making process at high threat (**Figures 5B, 5D**, and **5F**). In line with experimental data, threat gain varied substantially across reward conditions (1.2, 1.7, and 1.9 for no reward, water, and sucrose, respectively), exerting an effect opposite to that of drift rate and effectively canceling and reversing its influence.

As summarized in **Figure 5G**, if decisions are based solely on threat intensity, the likelihood of defensive decisions increases with increasing threat. If based solely on reward value, the likelihood of defensive decisions decreases with increasing reward. If only vigilance is considered, the likelihood of defensive decisions increases with both threat and reward. Our experimental findings and model simulations indicate that while threat intensity plays a central role in innate decision-making, the impact of reward on the decision-making process depends on the threat level. Under low-threat conditions, decisions are primarily driven by reward value, whereas under high-threat conditions, vigilance becomes the dominant factor.

3 Discussion

When confronted by a predator, an animal must decide whether to escape, freeze, or ignore the threat. This life-or-death decision is crucial for survival in the natural environment. In the laboratory, mice make similar decisions when exposed to an overhead expanding dark disc, which mimics the approach of an aerial predator. At first glance, their reactions may seem like simple startle responses to a sudden stimulus, such as thunder. However, a closer examination reveals that their behavior follows complex and structured patterns, distinct from simple reflexes. A growing body of research has demonstrated that these decisions involve cognitive control. Expanding on this idea, we show that an animal's response to a visual threat is further influenced by economic and social factors.

3.1 Main findings

We designed a behavioral paradigm to simulate how mice respond to visual threats during foraging (**Figure 1**). Quantitative analysis identified four distinct behavioral decisions in response to the looming stimulus (**Figure 1E**), characterized by stationary time, latency to flee, escape speed, hiding latency, and recovery time (**Figures 1G–1L**). Mice rapidly adapted to looming threats starting from the second trial (**Figure 2**). When a reward was introduced, decision-making was modulated by both threat intensity and reward value, albeit in different ways (**Figure 3**). Specifically, as threat intensity increased, mice shifted towards more defensive behaviors. In contrast, reward magnitude had a differential effect: at low-threat levels, higher rewards suppressed defensive responses, aligning with value-based decision theory; whereas at high-threat levels, greater rewards promoted defensive decisions, potentially due to heightened vigilance. Furthermore, innate decision-making was shaped by social hierarchy (**Figure 4**): dominant mice exhibited a stronger tendency towards defensive behaviors, while subordinates were reward-driven and less likely to flee. Finally, we proposed a drift-diffusion leaky integrator model to simulate the decision-making process, effectively capturing the key characteristics of the observed decision patterns (**Figure 5**).

3.2 Relation to earlier work

Aerial predators pose a significant threat to rodents. Previous studies have shown that overhead visual stimuli in the laboratory elicit robust defensive behaviors in rodents (Wallace et al., 2013; Yilmaz and Meister, 2013). Because such stimuli may mimic various predatory actions, the corresponding behavioral responses depend on specific physical properties of the stimulus. For example, an overhead expanding dark disc that mimics the approach of an aerial predator triggers more flight than freezing behavior (Yilmaz and Meister, 2013), while a small black moving dot that mimics a cruising predator induces more freezing behavior (De Franceschi et al., 2016). These findings suggest that the reaction is not a simple reflex but involves a decision-making process. Additional stimulus properties that influence the action selection include contrast, speed, size, and shape (De Franceschi et al., 2016; Evans et al., 2018; Yang et al., 2020). Support for a decision-making process also comes from its modulation by environmental factors and experience: mice freeze more when no refuge is available (Wei et al., 2015) and quickly learn that the looming stimulus poses no real threat (Lenzi et al., 2022; Vale et al., 2017; Zhong et al., 2023). This innate decision-making process is observed across invertebrates and vertebrates (Evans et al., 2019).

Compared with earlier work, the present study differs in two aspects. First, instead of manually annotating behavioral responses, we employed a machine-learning approach to classify behavioral decisions. This approach largely reduced the labor required for labeling and minimized misclassifications due to inter-individual variability. Second, unlike previous studies using 2D arenas, we designed a 1D linear arena to simulate foraging conditions, where the nest and reward zone were separated by a long corridor. Decision patterns observed here were similar to those reported in 2D environments (Yang et al., 2020). Consistent with previous findings (Lenzi et al., 2022), mice quickly learned that the looming stimulus was non-threatening starting from the 2nd trial. Interestingly, peak escape speed increased over the first five trials, indicating a brief period of heightened vigilance.

One potential concern with the linear arena design is that the looming stimulus is typically displayed at the end of the corridor, raising the possibility that mice may simply return to the nest upon reaching a physical boundary. To address this, we presented the stimulus halfway along the corridor. Even under this condition, mice quickly fled towards the nest – directly towards the threat – rather than away from it. This behavior strongly suggests that their escape responses are not simple reflexes, but instead incorporate the relative safety of the nest into their decisions (Figure S2).

3.3 Economic influence on decision-making

Value-based decision-making has been extensively studied across species (Rangel et al., 2008), including both behavioral paradigms and their underlying neural basis. In humans, decision-making under risk is well characterized by prospect theory, proposed by Daniel Kahneman and Amos Tversky nearly half a century ago (Kahneman and Tversky, 1979). This theory also accounts for the decision-making in rodents after extensive training (Constantinople et al., 2019). Our study demonstrates that mice make value-based decisions even in innate behaviors that require no training. A core component in value-based decision-making is the ability to learn the value of action outcomes. This learning process was clearly observed in our study: just within a few trials, mice habituated to the looming stimulus after recognizing that it posed no real harm (Figure 2). Their decisions reflected the relative values of reward and threat (Figure 3). Interestingly, while mice consistently exhibited value-based responses to threat, reward value-based decision-making was observed only under low-threat conditions. At high threat levels, however, mice exhibited more direct flight behavior regardless of reward value, indicating that their decisions were not value-based. At first glance, this may appear counter-intuitive. However, it can be understood through the lens of vigilance. In nature, foraging often coincides with

exposure to predation risk, and evolutionary pressure has favored prey that maintain heightened vigilance during foraging. As a result, reward value, threat intensity, and vigilance level are all correlated. Under high-threat conditions, the influence of reward value is suppressed because survival becomes the top priority and vigilance dominates the decisions. In other words, decisions at low threat levels are more rational, incorporating reward value with limited influence from threat, whereas decisions at high threat levels are more instinctive, dominated by the immediate need to escape.

What neural circuits underlie this innate decision-making under threat and reward? A growing body of research has implicated the superior colliculus (SC) as a central hub for processing looming-evoked defensive behaviors (Evans et al., 2018 [↗](#); Shang et al., 2015 [↗](#); Wei et al., 2015 [↗](#)). In parallel, reward-related signals have been found in several brain regions, including the ventral tegmental area (VTA) (Cohen et al., 2012 [↗](#); Schultz, 1998 [↗](#)), ventral striatum (Schultz et al., 1992 [↗](#)), orbitofrontal cortex (Tremblay and Schultz, 2000 [↗](#)), dorsal raphe nucleus (Liu et al., 2014 [↗](#); Miyazaki et al., 2011 [↗](#)), and cerebellum (Wagner et al., 2017 [↗](#)). Recent work has reported reward-modulated responses in the superficial SC of trained mice (Baruchin et al., 2022 [↗](#)). However, such responses may reflect brain-wide synchronized activity induced by overtraining (Steinmetz et al., 2019 [↗](#)), raising questions about their presence in untrained, natural behavior. Additionally, the lateral SC is involved in approach behaviors towards rewarding stimuli such as food or prey (Comoli et al., 2012 [↗](#); Krauzlis et al., 2013 [↗](#)), and the deep layers of SC directly project to dopaminergic neurons in the substantia nigra (Comoli et al., 2003 [↗](#)). Thus, it would be interesting to investigate how SC integrates reward and threat signals during innate decision-making.

3.4 Social influence on decision-making

Mice live in social groups in the wild, and social hierarchy plays a crucial role in shaping their behavior. The heightened vigilance and risk-averse behavior observed in dominant mice have also been observed in other rodents, such as voles (Kleiman et al., 2014 [↗](#)). From an evolutionary perspective, this behavioral strategy benefits not only individual survival but also the success of the species. Interestingly, the faster and more consistent escape responses observed in dominant mice are reminiscent of agonistic behaviors in crayfish (Krasne et al., 1997 [↗](#)). During social confrontations, dominant crayfish exhibit short-latency escape responses, while subordinates display more flexible escape behavior with longer and more variable latencies.

The neural mechanisms underlying status-dependent defensive decision-making remain poorly understood. In primates, serotonin promotes the acquisition of dominance, particularly under unstable social conditions (Raleigh et al., 1991 [↗](#)). Similarly, in crayfish, serotonin modulates escape behavior in a status-dependent manner, enhancing escape responses in dominant individuals while suppressing them in subordinates (Edwards and Kravitz, 1997 [↗](#); Yeh et al., 1997 [↗](#)). Investigating the neural circuit mechanisms driving status-dependent defensive decisions in mice would be an exciting avenue of research.

3.5 Mathematic modeling

The drift-diffusion leaky integrator model proposed builds on an integrator model of internal state (Gibson et al., 2015 [↗](#)) and incorporates a reward-driven drift-diffusion process (Ratcliff, 1978 [↗](#)). Conceptually, the integrator part in our model aligns with Lorenz's "hydraulic" model of motivation, which describes how internal drives shape behavior (Lorenz, 1950 [↗](#)). While our model resembles the leaky integrator used to model escape behavior in flies (Gibson et al., 2015 [↗](#)), it differs in several ways. First, the earlier model lacks a reward-related drift component. Second, while they modeled the looming effect as a delta function, our model allows the evidence level to vary continuously with the stimulus size. Third, the influence of vigilance is absent in that model. A similar model has been proposed (Evans et al., 2018 [↗](#)), but it did not incorporate reward and vigilance. Note that the evidence level for escape in our model is not equivalent to the fear

level (Anderson and Adolphs, 2014 [↗](#)); rather, when it crosses a threshold, the animal may enter a fear state. After initiating escape, the fear level decreases while the evidence level remains unaffected.

This hybrid model fits our experimental data well. Below we briefly discuss the roles of different parameters. The leakage rate α represents a drive towards the resting state. The perceived threat was modeled as the product of the habituation parameter h_i , the threat gain β , and the sensory input $s(t)$, where β is modulated by vigilance. For simplicity, h_i is fixed across all conditions. Under low-threat conditions, the relatively unchanged β across reward conditions suggests minimal variation in vigilance. In contrast, β varies with reward value under high-threat conditions, indicating that reward-related vigilance plays an important role in decision-making. The drift rate captures the influence of reward on decision-making. Interestingly, although the drift rate differences across reward conditions are small, they are sufficient to account for observed behavioral variability. Lastly, the diffusion rate δ plays a critical role, especially under low-threat conditions where the average evidence trajectory does not reach the threshold, which may reflect the fluctuation of the animal's internal state. The diffusion rate captures the trial-to-trial variability: even small noise can accumulate over time, resulting in large differences across individual trials and contributing to variability in escape latency. Overall, this computational framework not only accounts for our experimental observations but also offers a quantitative foundation for studying decision-making across species and ecological contexts. Understanding how these parameters are implemented at the circuit level is an exciting direction for future research.

4 Resource availability

4.1 Lead Contact

Further information and requests should be directed to and will be fulfilled by the Lead Contact: Ya-tang Li at yatangli@cibr.ac.cn.

4.2 Materials Availability

This study did not generate new unique reagents.

4.3 Data and Code Availability

Data and code will be available in a public repository upon acceptance of the manuscript.

5 Methods

5.1 Animal

Male C57BL/6J mice were group-housed on a 12-hour light / 12-hour dark cycle and used at ages 2-3 months. Behavioral experiments were conducted during the light phase. All experimental procedures were performed under the animal welfare guidelines and approved by the Institutional Animal Care and Use Committee at the Chinese Institute for Brain Research, Beijing.

5.2 Behavioral platform

The behavioral platform consisted of a nest, a linear arena, a radio frequency identification (RFID) system, a real-time mouse position detection system, and a reward delivery port. As shown in **Figure 1A** [↗](#), the nest was 40 cm (L) × 20 cm (W) × 30 cm (H), and the linear arena was 100 cm long × 10 cm wide × 30 cm high. They were connected by a one-way tunnel (20 cm × 3 cm × 3 cm) for

entering the arena and a 20 cm × 5 cm × 30 cm safe zone with a 5-cm × 5-cm one-way door for returning to the nest. Mouse identity was labeled via implanted RFID tags, which were detected by an RFID reader placed around the tunnel. Only one mouse was allowed to enter the arena at a time. Mice were rewarded at the end of the arena with either water or sucrose. Licking time and reward volume were recorded.

To monitor mouse position in real time, an OpenMV camera with a lens of 90° field of view was mounted on the ground, 110 cm beneath the arena. The mouse position was used to control the tunnel doors and to trigger stimulus presentation when the mouse entered the arena. Specifically, the tunnel had two doors: the first connected the tunnel to the nest, and the second door connected the tunnel to the arena. Once a mouse entered the tunnel, the first door closed and the second door opened. A second camera (LBAS-U350-74M) with a lens (FA0615A) was also placed on the ground to record animal behavior at 30 frames per second.

5.3 Visual stimulation

Looming stimuli were presented on a 55-inch monitor (121 cm × 68 cm) suspended 32 cm above the arena. Visual stimulation was implemented using Psychopy in Python and was aligned to the mouse's real-time location. The stimulus was a black disc that expanded ten times on a gray background ($\sim 65 \text{ cd/cm}^2$). For each time, the disc expanded from 0° to 20° of visual angle at a speed of 40°/s, followed by a stationary phase lasting 0.3 s. The inter-stimulus interval was randomized between 1 and 2 minutes. Two stimulus contrasts at 0.2 and 0.99 were displayed by adjusting the luminance of the disc.

5.4 Economical effects on the decision-making

All mice were habituated to the behavioral platform for two days before the looming test. On the first day, five mice from the same home cage were placed in the nest for 30 minutes. Each mouse was allowed to explore the arena for at least 10 minutes, twice. Afterward, all five mice were returned to the nest with all gates open, allowing free exploration of the arena for two hours to ensure that each mouse learned a reward was available at the end of the linear arena. On the second day, each mouse was given one hour to explore the arena and further acclimate to the environment. The looming test was conducted the following day. An overhead looming stimulus with either 0.25 or 0.99 contrast was displayed when a mouse's position was detected within 20 cm of the arena's end. For each mouse, both the looming contrast and reward type were fixed. In total, we collected and analyzed behavioral data from 259 trials across 29 mice.

To control for individual variation in decision-making, we compared the same animal's behavior under reward and no-reward conditions. Two groups of mice were used in the experiment. In the first group ($n = 5$), mice were first tested under the reward condition followed by the no-reward condition. In the second group ($n = 4$), the order was reversed. For the reward condition, mice were water-deprived one day before the exploration, and water was provided via the reward port. In the non-reward condition, mice were not water-deprived, and the reward port was removed. Before the looming test, mice were acclimated to the linear arena for two days as described above. On the third day, a looming stimulus with 0.25 contrast was displayed for five trials. In total, behavioral data from 84 trials across nine mice were recorded and analyzed.

5.4.1 Social effects on the decision-making

To investigate the impact of social hierarchy on decision-making, we first determined the social rank of each mouse pair using the tube test (Wang et al., 2011 [DOI](#)). Mice were co-housed with a glass tube (3 cm diameter, 10 cm long) for one week, then trained to cross a 30 cm tube 10 times per day over two consecutive days. On the third day, the tube test was conducted up to 6 times until the rank order was stable for 4 consecutive trials. Pairs that failed to reach a stable hierarchy were excluded (1 out of 6 pairs).

Prior to the looming test, mice were water-deprived and allowed to explore the linear arena for two hours per day over three days, during which water rewards were delivered at the end of the arena. Over the following two days, the looming test with a stimulus contrast of 0.99 was performed for two hours per day. After the looming test, mice continued to explore the arena for an additional two days, after which their social rank was reassessed with the tube test. All remaining pairs maintained a stable rank order.

To assess whether the tube test itself influenced defensive decision-making, additional looming experiments were conducted on four mouse pairs before and after introducing the tube test. Specifically, the first looming experiment was conducted prior to the tube test in [Figure 4A](#). The behavioral patterns in a total of 180 trials from 18 mice were recorded and analyzed.

5.5 Behavioral quantification

Animal behaviors in these trials were quantified in three steps. First, two key points – the nose and tail base – were labeled and tracked in the recorded videos using DeepLabCut ([Figure S1A](#)). Second, individual 30-second trials were extracted, each consisting of 10 seconds before, 8 seconds during, and 12 seconds after the looming stimulus. For each trial, 19 behavioral features were defined from the tracked key points. Finally, these features were fed into a random forest model with a maximum depth of 5 to classify animal behaviors. The model was trained with 238 trials and tested in 102 trials. The accuracy score for the test set is 0.95. In total, behavioral decisions in 3862 trials from 140 mice were classified into four types: direct flight, flight after assessment, freezing, and no reaction ([Figure S1B](#)).

5.6 Behavioral modeling

The animal's decision-making process in response to the looming stimulus during foraging was modeled as a drift-diffusion leaky integrator. An escape response is triggered when the accumulated evidence level crosses a defined threshold:

$$\frac{dx_i}{dt} = -\alpha \cdot x_i(t) + h_i \cdot \beta \cdot s(t) - r + \delta \cdot \frac{dW}{dt} \quad (1)$$

$$E(t) = \mathcal{H}(x_i(t) - x_{thr}) \quad (2)$$

Here, $x_i(t)$ is the evidence level for escape at time t on the i th trial. The parameter α is the leakage rate, which drives the evidence level towards zero and is the reciprocal of the integration time constant. The stimulus function $s(t)$ denotes the normalized diameter of the looming stimulus, while β is the threat gain that modulates the perceived threat level based on stimulus contrast and the animal's vigilance. The habituation parameter h_i captures the adaptation to repeated stimuli. The term r denotes the drift rate reflecting the perceived reward value. $W(t)$ is a Wiener process, such that $W(t + \Delta t) - W(t) \sim \mathcal{N}(0, \Delta t)$; δ is the diffusion rate. The binary variable $E(t)$ indicates the escape decision, with $\mathcal{H}$ denoting the Heaviside step function.

Given that animals exhibited rapid habituation to the looming stimulus for the first five trials followed by a slower decline, the habituation effect was modeled as the average of fast and slow exponential decay components:

$$h_i = (e^{-tr_i/\tau_f} + e^{-tr_i/\tau_s})/2 \quad (3)$$

where tr_i is the i th trial; τ_f and τ_s are the time constants for the fast and slow components, respectively.

Model parameters were fitted using the lmfit Python package. In [Equation 1](#), $\alpha = 1.78$ and $\delta = 6.9$. For low-threat conditions, $\beta = 1, 1, 0.9$ for no reward, water, and sucrose, respectively. For high-threat conditions, $\beta = 1.8, 2.4, 2.5$ across reward conditions. The drift rate $r = 0, 0.02, 0.07$ for

no reward, water, and sucrose, respectively. In [Equation 2](#), the decision threshold $x_{thr} = 0.77$. For the habituation function in [Equation 3](#), $\tau_f = 300$ and $\tau_s = 7.96$.

5.7 Quantification and statistical analysis

No statistical method was used to predetermine sample size. The Shapiro-Wilk test was applied to assess the normality of data distributions. For comparison between two groups, a two-sample t -test or paired t -test was used for normally distributed data; otherwise, non-parametric tests, including the Mann–Whitney U-test and the one-sample Wilcoxon signed-rank test, was applied. Two-way analysis of variance (ANOVA) followed by post-hoc tests was used for multi-group comparisons. Detailed statistical information for each experiment is provided in the Results section and figure legends.

6 Supplemental information

Here we report details related to the Results and Methods sections.

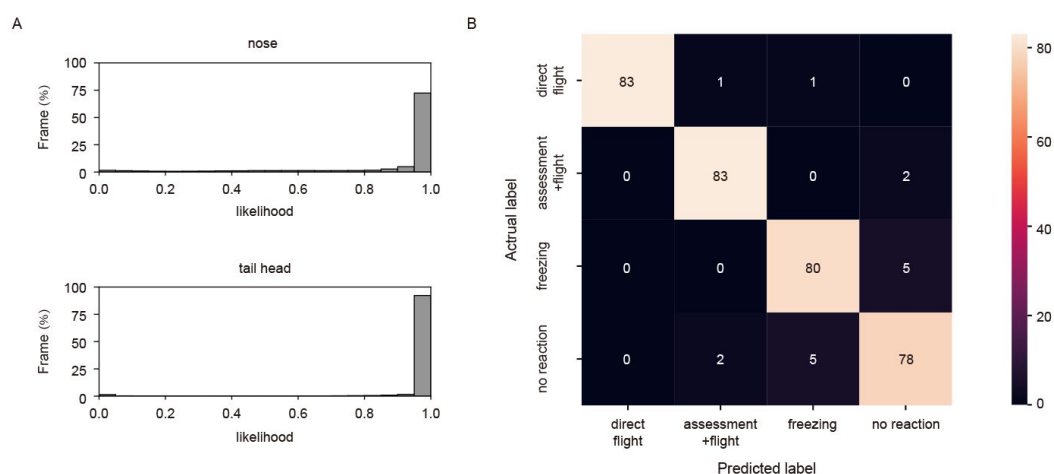


Figure S1

Quality control for behavioral classification model.

(A) Tracking accuracy of mouse nose and tail base in DeepLabCut. (B) Model performance evaluated by a confusion matrix on the test dataset.

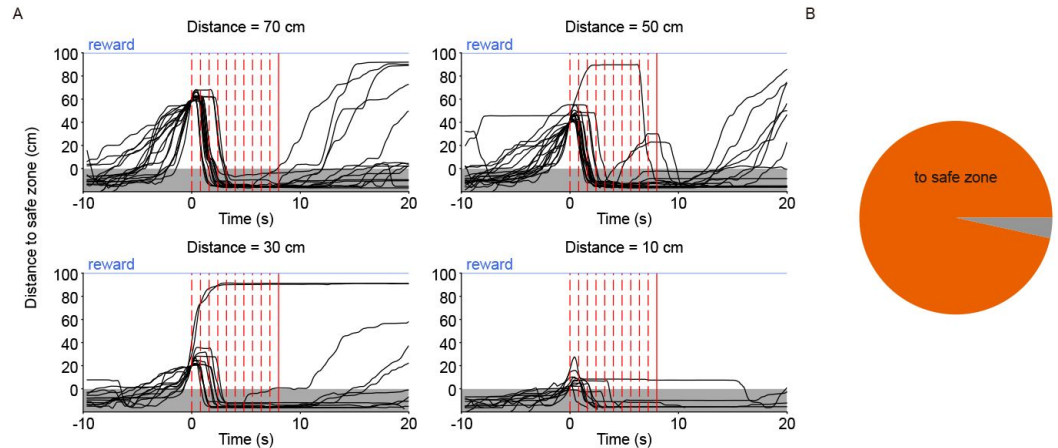


Figure S2

Behavioral responses to looming stimuli presented at four distances from the safe zone.

(A) Distance to the safe zone. Red dashed lines mark the onset of each stimulus repetition; solid lines mark the end of the last repetition. Grey shade indicates the safe zone. N = 18 (70 cm), 19 (50 cm), 13 (30 cm), and 11 (10 cm) trials. (B) Distribution of escape directions.

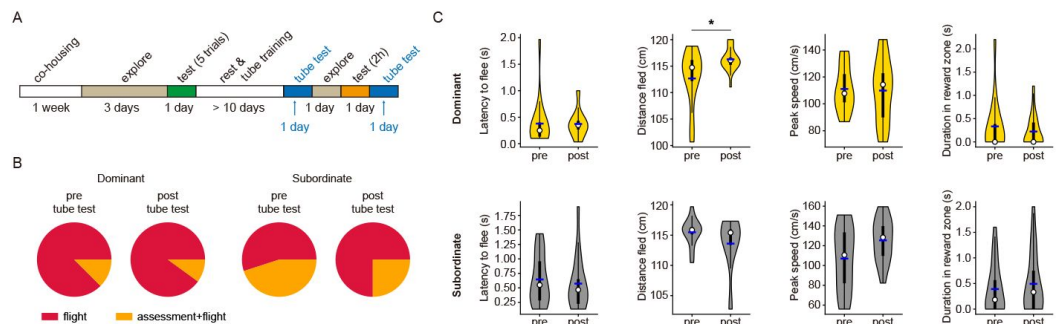


Figure S3

Comparison of behavioral responses to looming stimuli before and after the tube test.

(A) Schematic timeline of the looming experiment before and after the tube test. (B) Behavioral decisions for dominant and subordinate mice before and after the tube test; Chi-squared test. (C) Violin plots showing latency to flee, distance fled, peak fleeing speed, and duration in the reward zone for dominant (top) and subordinate (bottom) mice before and after the tube test; Mann-Whitney-Wilcoxon test. * $p < 0.05$.

Acknowledgements

We thank Lei Zhang, Xueting Sun, Haojun Sang, Qun Zhang, Bing Zhao, and Zhuofan Li in the CIBR instrumentation core for their help in designing and building the linear arena. Ya-tang Li is supported by the National Natural Science Foundation of China (32271060), the Natural Science Foundation of Beijing Municipality (IS23073), and the start-up fund from CIBR. Ling-yun Li is supported by the Natural Science Foundation of Beijing Municipality (5244028), the National Natural Science Foundation of China (32471071), and the R&D Program of Beijing Municipal Education Commission (1240030201).

Additional information

Author contributions

Ya-tang Li supervised the project; Ya-tang Li, Zhe Li, and Yidan Sun designed the experiments; Zhe Li collected all the data; Zhe Li, Jiahui Wang, and Jialin Li analyzed the data; Ya-tang Li carried out the neural modeling; Zhe Li and Ya-tang Li prepared figures; Ya-tang Li, Ling-yun Li, and Zhe Li wrote the manuscript.

References

- Anderson D. J., Adolphs R. (2014) **A Framework for Studying Emotions across Species** *Cell* **157**:187–200 [Google Scholar](#)
- Baruchin L. J., Alleman M., Schröder S. (2022) **Reward modulates visual responses in the superficial superior colliculus of mice** [Google Scholar](#)
- Buck L. (1966) **Reaction time as a measure of perceptual vigilance** *Psychological Bulletin US: American Psychological Association* **65**:291–304 [Google Scholar](#)
- Burgess C. P., Lak A., Steinmetz N. A., Zatzka-Haas P., Reddy C. B., Jacobs E. A. K., Linden J. F., Paton J. J., Ranson A., Schröder S., Soares S., Wells M. J., Wool L. E., Harris K. D., Carandini M. (2017) **High-Yield Methods for Accurate Two-Alternative Visual Psychophysics in Head-Fixed Mice** *Cell Reports Elsevier* **20**:2513–2524 [Google Scholar](#)
- Caraco T., Martindale S., Pulliam H. R. (1980) **Avian flocking in the presence of a predator** *Nature Nature Publishing Group* **285**:400–401 [Google Scholar](#)
- Cohen J. Y., Haesler S., Vong L., Lowell B. B., Uchida N. (2012) **Neuron-type-specific signals for reward and punishment in the ventral tegmental area** *Nature Nature Publishing Group* **482**:85–88 [Google Scholar](#)
- Comoli E., Coizet V., Boyes J., Bolam J. P., Canteras N. S., Quirk R. H., Overton P. G., Redgrave P. (2003) **A direct projection from superior colliculus to substantia nigra for detecting salient visual events** *Nature Neuroscience* **6**:974–980 [Google Scholar](#)
- Comoli E., Das Neves Favaro P., Vautrelle N., Leriche M., Overton P. G., Redgrave P. (2012) **Segregated Anatomical Input to Sub-Regions of the Rodent Superior Colliculus Associated with Approach and Defense** *Frontiers in Neuroanatomy* **6** [Google Scholar](#)
- Constantinople C. M., Piet A. T., Brody C. D. (2019) **An Analysis of Decision under Risk in Rats** *Current Biology* **29**:2066–2074 [Google Scholar](#)
- De Franceschi G., Vivattanasarn T., Saleem A., Solomon S. (2016) **Vision Guides Selection of Freeze or Flight Defense Strategies in Mice** *Current Biology* **26**:2150–2154 [Google Scholar](#)
- Edwards D. H., Kravitz E. A. (1997) **Serotonin, social status and aggression** *Current Opinion in Neurobiology* **7**:812–819 [Google Scholar](#)
- Evans D. A., Stempel A. V., Vale R., Branco T. (2019) **Cognitive Control of Escape Behaviour** *Trends in Cognitive Sciences Elsevier* **23**:334–348 [Google Scholar](#)
- Evans D. A., Stempel A. V., Vale R., Ruehle S., Lefler Y., Branco T. (2018) **A synaptic threshold mechanism for computing escape decisions** *Nature* **558**:590–594 [Google Scholar](#)
- Fanselow M. S., Lester L. S. (1988) **A functional behavioristic approach to aversively motivated behavior: predatory imminence as a determinant of the topography of defensive behavior** In: Bolles R. C., Beecher M. D., editors. *Evolution and Learning* Lawrence Erlbaum Associates pp. 185–212 [Google Scholar](#)

Gibson W. T., Gonzalez C. R., Fernandez C., Ramasamy L., Tabachnik T., Du R. R., Felsen P. D., Maire M. R., Perona P., Anderson D. J. (2015) **Behavioral Responses to a Repetitive Visual Threat Stimulus Express a Persistent State of Defensive Arousal in *Drosophila*** *Current Biology* **25**:1401–1415 [Google Scholar](#)

Kahneman D., Tversky A. (1979) **Prospect Theory: An Analysis of Decision under Risk** *Econometrica* Wiley, Econometric Society] **47**:263–291 [Google Scholar](#)

Kleiman M., Bodek S., Eilam D. (2014) **Who are the bosses? Group influence on the behavior of voles following owl attack** *Behavioural Processes* **108**:183–190 [Google Scholar](#)

Krasne F. B., Shamsian A., Kulkarni R. (1997) **Altered Excitability of the Crayfish Lateral Giant Escape Reflex during Agonistic Encounters** *Journal of Neuroscience* **17**:709–716 [Google Scholar](#)

Krauzlis R. J., Lovejoy L. P., Zénon A. (2013) **Superior Colliculus and Visual Spatial Attention** *Annual Review of Neuroscience* **36**:165–182 [Google Scholar](#)

Lenzi S. C., Cossell L., Grainger B., Olesen S. F., Branco T., Margrie T. W. (2022) **Threat history controls flexible escape behavior in mice** *Current Biology* **32**:2972–2979 [Google Scholar](#)

Liden W. H., Herberholz J. (2008) **Behavioral and neural responses of juvenile crayfish to moving shadows** *The Journal of Experimental Biology* **211**:1355–1361 [Google Scholar](#)

Liden W. H., Phillips M. L., Herberholz J. (2010) **Neural control of behavioural choice in juvenile crayfish** *Proceedings of the Royal Society B: Biological Sciences* **277**:3493–3500 [Google Scholar](#)

Liu Z., Zhou J., Li Y., Hu F., Lu Y., Ma M., Feng Q., Zhang J.-E., Wang D., Zeng J., Bao J., Kim J.-Y., Chen Z.-F., El Mestikawy S., Luo M. (2014) **Dorsal raphe neurons signal reward through 5-HT and glutamate** *Neuron* **81**:1360–1374 [Google Scholar](#)

Lorenz K. Z. (1950) **The comparative method in studying innate behavior patterns** *Physiological mechanisms in animal behavior. (Society's Symposium IV.)* Oxford, England: Academic Press :221–268 [Google Scholar](#)

Mathis A., Mamidanna P., Cury K. M., Abe T., Murthy V. N., Mathis M. W., Bethge M. (2018) **DeepLabCut: markerless pose estimation of user-defined body parts with deep learning** *Nature Neuroscience* **21**:1281–1289 [Google Scholar](#)

Meister M. (2022) **Learning, fast and slow** *Current Opinion in Neurobiology* **75**:102555 [Google Scholar](#)

Miyazaki K., Miyazaki K. W., Doya K. (2011) **Activation of dorsal raphe serotonin neurons underlies waiting for delayed rewards** *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience* **31**:469–479 [Google Scholar](#)

Raleigh M. J., McGuire M. T., Brammer G. L., Pollack D. B., Yuwiler A. (1991) **Serotonergic mechanisms promote dominance acquisition in adult male vervet monkeys** *Brain Research* **559**:181–190 [Google Scholar](#)

- Rangel A., Camerer C., Montague P. R. (2008) **A framework for studying the neurobiology of value-based decision making** *Nature Reviews Neuroscience* **9**:545–556 [Google Scholar](#)
- Ratcliff R. (1978) **A theory of memory retrieval** *Psychological Review* **85**:59–108 [Google Scholar](#)
- Schadegg A. C., Herberholz J. (2017) **Satiation level affects anti-predatory decisions in foraging juvenile crayfish** *Journal of Comparative Physiology. A, Neuroethology, Sensory, Neural, and Behavioral Physiology* **203**:223–232 [Google Scholar](#)
- Schultz W. (1998) **Predictive reward signal of dopamine neurons** *Journal of Neurophysiology* **80**:1–27 [Google Scholar](#)
- Schultz W., Apicella P., Scarnati E., Ljungberg T. (1992) **Neuronal activity in monkey ventral striatum related to the expectation of reward** *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience* **12**:4595–4610 [Google Scholar](#)
- Shang C., Liu Z., Chen Z., Shi Y., Wang Q., Liu S., Li D., Cao P. (2015) **A parvalbumin-positive excitatory visual pathway to trigger fear responses in mice** *Science* **348**:1472–1477 [Google Scholar](#)
- Steinmetz N. A., Zatka-Haas P., Carandini M., Harris K. D. (2019) **Distributed coding of choice, action and engagement across the mouse brain** *Nature* Nature Publishing Group **576**:266–273 [Google Scholar](#)
- Tammero L. F., Dickinson M. H. (2002) **Collision-avoidance and landing responses are mediated by separate pathways in the fruit fly, *Drosophila melanogaster*** *The Journal of Experimental Biology* **205**:2785–2798 [Google Scholar](#)
- Tremblay L., Schultz W. (2000) **Reward-related neuronal activity during go-nogo task performance in primate orbitofrontal cortex** *Journal of Neurophysiology* **83**:1864–1876 [Google Scholar](#)
- Vale R., Evans D. A., Branco T. (2017) **Rapid Spatial Learning Controls Instinctive Defensive Behavior in Mice** *Current Biology* **27**:1342–1349 [Google Scholar](#)
- Wagner M. J., Kim T. H., Savall J., Schnitzer M. J., Luo L. (2017) **Cerebellar granule cells encode the expectation of reward** *Nature* **544**:96–100 [Google Scholar](#)
- Wallace D. J., Greenberg D. S., Sawinski J., Rulla S., Notaro G., Kerr J. N. D. (2013) **Rats maintain an overhead binocular field at the expense of constant fusion** *Nature* **498**:65–69 [Google Scholar](#)
- Wang F., Zhu J., Zhu H., Zhang Q., Lin Z., Hu H. (2011) **Bidirectional Control of Social Hierarchy by Synaptic Efficacy in Medial Prefrontal Cortex** *Science* American Association for the Advancement of Science **334**:693–697 [Google Scholar](#)
- Wei P., Liu N., Zhang Z., Liu X., Tang Y., He X., Wu B., Zhou Z., Liu Y., Li J., Zhang Y., Zhou X., Xu L., Chen L., Bi G., Hu X., Xu F., Wang L. (2015) **Processing of visually evoked innate fear by a non-canonical thalamic pathway** *Nature Communications* **6**:6756 [Google Scholar](#)
- Yang X., Liu Q., Zhong J., Song R., Zhang L., Wang L. (2020) **A simple threat-detection strategy in mice** *BMC Biology* **18**:93 [Google Scholar](#)

Ydenberg R. C., Dill L. M. (1986) **The Economics of Fleeing from Predators** In: Rosenblatt J. S., Beer C., Busnel M.-C., Slater P. J. B., editors. *Advances in the Study of Behavior* Academic Press pp. 229–249 [Google Scholar](#)

Yeh S.-R., Musolf B. E., Edwards D. H. (1997) **Neuronal Adaptations to Changes in the Social Dominance Status of Crayfish** *Journal of Neuroscience Society for Neuroscience Section: Articles* **17**:697–708 [Google Scholar](#)

Yilmaz M., Meister M. (2013) **Rapid Innate Defensive Responses of Mice to Looming Visual Stimuli** *Current Biology* **23**:2011–2015 [Google Scholar](#)

Zhong J., Liu Q., Yang X., Su J., Li X., Luo M., Wang L. (2023) **Mice learn from the predator-attack experience to accelerate flight behavior via optimizing the strategy of environment exploration** *Biochemical and Biophysical Research Communications* **665**:26–34 [Google Scholar](#)

Author information

Zhe Li

Academy for Advanced Interdisciplinary Studies, Peking University, Beijing, China, Beijing Institute for Brain Research, Chinese Academy of Medical Sciences & Peking Union Medical College, Beijing, China, Chinese Institute for Brain Research, Beijing, China

Jiahui Wang

Chinese Institute for Brain Research, Beijing, China

Yidan Sun

Chinese Institute for Brain Research, Beijing, China

Jialin Li

Chinese Institute for Brain Research, Beijing, China

Ling-yun Li

Department of Neurobiology, School of Basic Medical Sciences, Capital Medical University, Beijing, China

Ya-tang Li

Beijing Institute for Brain Research, Chinese Academy of Medical Sciences & Peking Union Medical College, Beijing, China, Chinese Institute for Brain Research, Beijing, China
ORCID iD: [0000-0003-2763-1534](https://orcid.org/0000-0003-2763-1534)

For correspondence: yatangli@cibr.ac.cn

Editors

Reviewing Editor

Alicia Izquierdo

University of California, Los Angeles, Los Angeles, United States of America

Senior Editor

Joshua Gold

University of Pennsylvania, Philadelphia, United States of America

Reviewer #1 (Public review):

Summary:

This study investigates how mice make defensive decisions when exposed to visual threats and how those decisions are influenced by reward value and social hierarchy. Using a naturalistic foraging setup and looming stimuli, the authors show that higher threat leads to faster escape, while lower threat allows mice to weigh reward value. Dominant mice behave more cautiously, showing higher vigilance. The behavioral findings are further supported by a computational model aimed at capturing how different factors shape decisions.

Strengths:

- (1) The behavioral paradigm is well-designed and ethologically relevant, capturing instinctive responses in a controlled setting.
- (2) The paper addresses an important question: how defensive behaviors are influenced by social and value-based factors.
- (3) The classification of behavioral responses using machine learning is a solid methodological choice that improves reproducibility.

Weaknesses:

- (1) Key parts of the methods are hard to follow, especially how trials are selected and whether learning across trials is fully controlled for. For example, it is unclear whether animals are in the nest during the looming stimulus presentations. The main text and methods should clarify whether multiple mice are in the nest simultaneously and whether only one mouse is in the arena during looming exposure. From the description, it seems that all mice may be freely exploring during some phases, but only one is allowed in the arena at a time during stimulus presentation. This point is important for understanding the social context and potential interactions, and should be clearly explained in both the main text and methods.
- (2) It is often unclear whether the data shown (especially in the main summary figures) come from the first trial or are averages across several exposures. When is the cut-off for trials of each animal? How do we know how many trial presentations were considered, and how learning at different rates between individuals is taken into account when plotting all animals together? This is important because the looming stimulus is learned to be harmless very quickly, so the trial number strongly affects interpretation.
- (3) The reward-related effects are difficult to interpret without a clearer separation of learning vs first responses.
- (4) The model reproduces observed patterns but adds limited explanatory or predictive power. It does not integrate major findings like social hierarchy. Its impact would be greatly improved if the authors used it to predict outcomes under novel or intermediate conditions.
- (5) Some conclusions (e.g., about vigilance increasing with reward) are counterintuitive and need stronger support or alternative explanations. Regarding the interpretation of social differences in area coverage, it's also possible that the observed behavioral differences reflect access to the nesting space. Dominant mice may control the nest, forcing subordinates to remain in the open arena even during or after looming stimuli. In this case, subordinates may be choosing between the threat of the dominant mouse and the external visual threat. The current data do not distinguish between these possibilities, and the authors do not

provide evidence to support one interpretation over the other. Including this alternative explanation or providing data that addresses it would strengthen the conclusions.

(6) While potential neural circuits are mentioned in the discussion, an earlier introduction of candidate brain regions and their relevance to threat and value processing would help ground the study in existing systems neuroscience.

(7) Some figures are difficult to interpret without clearer trial/mouse labeling, and a few claims in the text are stronger than what the data fully support. Figure 3H is done for low contrast, but the interesting findings will be to do this experiment with high contrast. Figure 4H - I don't understand this part. If the amount of time in the center after the loom changes for subordinate mice, how does this lead to the conclusion that they spend most of their time in the reward zone?. Figure 3A - The example shown does not seem representative of the claim that high contrast stimuli are more likely to trigger escape. In particular, the 10% sucrose condition appears to show more arena visits under low contrast than high contrast, which seems to contradict that interpretation. Also, the plot currently uses trials on the Y-axis, but it would be more informative to show one line per animal, using only the first trial for each. This would help separate initial threat responses from learning effects and clarify individual variability.

(8) The analysis does not explore individual variability in behavior, which could be an important source of structure in the data. Without this, it is difficult to know whether social hierarchy alone explains behavioral differences or if other stable traits (e.g., anxiety level, prior experiences) also contribute.

(9) The study shows robust looming responses in group-housed animals, which contrasts with other studies that often require single housing to elicit reliable defensive responses. It would be valuable for the authors to discuss why their results differ in this regard and whether housing conditions might interact with social rank or habituation.

<https://doi.org/10.7554/eLife.107306.1.sa2>

Reviewer #2 (Public review):

Zhe Li and colleagues investigate how mice exposed to visual threats and rewards balance their decisions in favour of consuming rewards or engaging in defensive actions. By varying threat intensity and reward value, they first confirm previous findings showing that defensive responses increase with threat intensity and that there is habituation to the threat stimulus. They then find that water-deprived mice have a reduced probability of escaping from low contrast visual looming stimuli when water or sucrose are offered in the environment, but that when the stimulus contrast is high, the presence of sucrose or water increases the probability of escape. By analysing behaviour metrics such as the latency to flee from the threat stimulus, they suggest that this increase in threat sensitivity is due to increased vigilance. Analysis of this behaviour as a function of social hierarchy shows that dominant mice have higher threat sensitivity, which is also interpreted as being due to increased vigilance. These results are captured by a drift diffusion model variant that incorporates threat intensity and reward value.

The main contribution of this work is to quantify how the presence of water or sucrose in water-deprived mice affects escape behaviour. The differential effects of reward between the low and high contrast conditions are intriguing, but I find the interpretation that vigilance plays a major role in this process is not supported by the data. The idea that reward value exerts some form of graded modulation of the escape response is also not supported by the data. In addition, there is very limited methodological information, which makes assessing

the quality of some of the analyses difficult, and there is no quantification of the quality of the model fits.

(1) The main measure of vigilance in this work is reaction time. While reaction time can indeed be affected by vigilance, reaction times can vary as a function of many variables, and be different for the same level of vigilance. For example, a primate performing the random dot motion task exhibits differences in reaction times that can be explained entirely by the stimulus strength. Reaction time is therefore not a sound measure of vigilance, and if a goal of this work is to investigate this parameter, then it should be measured. There is some attempt at doing this for a subset of the data in Figure 3H, by looking at differences in the action of monitoring the visual field (presumably a rearing motion, though this is not described) between the first and second trials in the presence of sucrose. I find this an extremely contrived measure. What is the rationale for analysing only the difference between the first and second trials? Also, the results are only statistically significant because the first trial in the sucrose condition happens to have zero up action bouts, in contrast to all other conditions. I am afraid that the statistics are not solid here. When analysing the effects of dominance, a vigilance metric is the time spent in the reward zone. Why is this a measure of vigilance? More generally, measuring vigilance of threats in mice requires monitoring the position of the eyes, which previous work has shown is biased to the upper visual field, consistent with the threat ecology of rodents.

(2) In both low and high contrast conditions, there are differences in escape behaviour between no reward and water or sucrose presence, but no statistically significant differences between water and sucrose (eg, Figure 3B). I therefore find that statements about reward value are not supported by the data, which only show differences between the presence or absence of reward. Furthermore, there is a confound in these experiments, because according to the methods, mice in the no-reward condition were not water deprived. It is thus possible that the differences in behaviour arise from differences in the underlying state.

(3) There is very little methodological information on behavioural quantification. For example, what is hiding latency? Is this the same as reaction time? Time to reach the safe zone? What exactly is distance fled? I don't understand how this can vary between 20 and 100cm. Presumably, the 20cm flights don't reach the safe place, since the threat is roughly at the same location for each trial? How is the end of a flight determined? How is duration measured in reward zone measures, e.g., from when to when? How is fleeing onset determined?

(4) There is little methodological information on how the model was fit (for example, it is surprising that in the no reward condition, the r parameter is exactly 0. What this constrained in any way), and none of the fit parameters have uncertainty measures so it is not possible to assess whether there are actually any differences in parameters that are statistically significant.

<https://doi.org/10.7554/eLife.107306.1.sa1>

Reviewer #3 (Public review):

Male mice were tested in a classic behavioral "flee the looming stimulus" paradigm. This is a purely behavioral study; no neural analyses were done. Mice were housed socially, but faced the looming stimulus individually. Drift-diffusion modeling found that reward-level interacted with threat level such that at low-threat levels, reward contrasted with threat as classically expected (high reward overwhelms low threat, low threat overwhelms low reward), but that reward aligned with threat at higher threat levels.

Note that they define threat level by the darkness of the looming stimulus. I am not sure that darker stimuli are more threatening to mice. But maybe. Figure 3 shows that mice react more quickly to high contrast looming stimuli, but can the authors distinguish between the ability to detect the visual signal from considering it a more dangerous threat? (The fact that vigilance makes a difference in the high contrast condition, not the low contrast condition, actually supports the author's hypotheses here.)

The drift-diffusion model (DDM) is fine. I note that the authors included a "leakage rate", which is not a standard DDM parameter (although I like including it). I would have liked to see more about the parameters. What were the distributions? What did the parameters correlate with behaviorally? I would have liked to see distributions of the parameters under the different conditions and different animals. Figure 2C shows the progression of learning. How do the fit parameters change over time as mice shift from choice to choice? How do the parameters change over mice? How do the parameters change over distance to the threat/distance to safety (as per Fanselow and Lester 1988)? They did a supplemental experiment where the threat arrived halfway along the corridor - we could get a lot more detail about that experiment - how did it change the modeling?

Overall, this is a reasonable study showing mostly unsurprising results. I think the authors could do more to connect the vigilance question to their results (which seems somewhat new to me).

Although the data appear generally fine and the modeling reasonable, the authors do not do the necessary work to set themselves within the extensive literature on decision-making in mice retreating from threats.

First of all, this is not a new paradigm; variants of this paradigm have been used since at least the 1980s. There is an **extensive** literature on this, including extensive theoretical work on the relation of fear and other motivational factors. I recommend starting with the classic Fanselow and Lester 1988 paper (which they cite, but only in passing), and the reviews by Dean Mobbs and Jeansok Kim, and by Denis Paré and Greg Quirk, which have explicit theoretical proposals that the authors can compare their results to. I would also recommend that the authors look into the "active avoidance" literature. Moreover, to talk about a mouse running from a looming stimulus without addressing the other "flee the predator" tasks is to miss a huge space for understanding their results. Again, I would start with the reviews above, but also strongly urge the authors to look at the Robogator task (work by June-Seek Choi and Jeansok Kim, work by Denis Paré, and others).

Similarly, in their anatomical review, they do not mention the amygdala. Given the extensive literature on the role of the amygdala in retreating from danger, both in terms of active avoidance and in terms of encoding the danger itself, it would surprise me greatly if this behavior does not involve amygdala processing. (If there is evidence that the amygdala does not play a role here, but that the superior colliculus does, then that would be a **very** important result that needs to be folded into our understanding of decision-making systems and neural computational processing.)

Second, there is an extensive economic literature on non-human animals in general and on rodents in particular. Again, the authors seem unaware of this work, which would provide them with important data and theories to broaden the impact of their results (by placing them within the literature). First, there are explicit economic literatures in terms of positively-valenced conflicts (e.g., neuroeconomics within the primate literature, sequential foraging and delay-discounting tasks within the rodent literature), but also there is a long history within the rodent conditioning world, such as the classic work by Len Green and Peter Shizgal. I would strongly urge the authors to explore the motivational conflict literature

by people like Gavin McNally, Greg Quirk, and Mark Andermann. Again, putting their results into this literature will increase the impact of their experiment and modeling.

<https://doi.org/10.7554/eLife.107306.1.sa0>